THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A triangle has sides 11 cm, 25 cm and 30 cm. It is: 1

(a) not right-angled **(b)** equilateral **(c)** right-angled **(d)** cannot be said

2. The diagonal of a square of side 5 cm (in cm) is: 1

(a) 10 **(b)** $\sqrt{51}$ **(c)** 5 **(d)** $5\sqrt{2}$

3. The hypotenuse of a right triangle is 41 cm and one leg is 9 cm. Its area (in cm²) is: 1

(a) 189 **(b)** 360 **(c)** 369/2 **(d)** 180

4. If the sum of the squares of two sides of a triangle equals the square of the third side, then the triangle is: 1

(a) acute-angled **(b)** equilateral **(c)** right-angled **(d)** obtuse-angled

5. The legs of a right triangle are 3 cm and 5 cm. Its hypotenuse (in cm) is: 1

(a) $2\sqrt{2}$ **(b)** 8 **(c)** $\sqrt{34}$ **(d)** $\sqrt{35}$

6. The legs of a right triangle are 5 cm and 14 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 19 **(b)** 361 **(c)** 221 **(d)** 70

7. Taking $m = 10$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 20, 99, 101 **(b)** 10, 99, 101 **(c)** 20, 99, 102 **(d)** 20, 100, 101

8. A triangle has sides 24 cm, 45 cm and 51 cm. It is: 1

(a) equilateral **(b)** not right-angled **(c)** cannot be said **(d)** right-angled

9. The diagonal of a square of side 6 cm (in cm) is: 1

(a) 6 **(b)** $\sqrt{73}$ **(c)** 12 **(d)** $6\sqrt{2}$

10. The hypotenuse of a right triangle is 26 cm and one leg is 10 cm. Its area (in cm²) is: 1

(a) 121 **(b)** 130 **(c)** 240 **(d)** 120

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The sides of a triangle are 18 cm, 24 cm, 32 cm. Is it a right triangle? Which angle is the right angle? 2

12. Find the length of the diagonal of a square of side 8 cm. 2

13. Find the Pythagorean triplet with the smallest member 32. 2

14. The sides of a triangle are 45 cm, 108 cm, 117 cm. Is it a right triangle? Which angle is the right angle? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The hypotenuse of an isosceles right triangle is $7\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

16. A ladder 26 m long reaches a window 24 m above the ground. Find the distance of the foot of the ladder from the wall. 3

17. The diagonals of a rhombus are 22 cm and 120 cm. Find its side. 3

OR

The hypotenuse of an isosceles right triangle is $9\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

18. A ladder 37 m long reaches a window 35 m above the ground. Find the distance of the foot of the ladder from the wall. 3

OR

The diagonals of a rhombus are 32 cm and 60 cm. Find its side. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The diagonals of a rhombus are 20 cm and 48 cm. Find its side, perimeter and area. 5

OR

Two poles of heights 17 m and 5 m stand on level ground 16 m apart. Find the distance between their tops. 5

20. The diagonals of a rhombus are 24 cm and 70 cm. Find its side, perimeter and area. 5

OR

Two poles of heights 45 m and 5 m stand on level ground 42 m apart. Find the distance between their tops. 5

END OF QUESTION PAPER — Total: 40 Marks

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